

tests/dynamic/analyzers/ run_analyzer_selftests.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active §11.4.107(10) self-validation runner for all six dynamic
data-plane analyzers. **Runnable today** against bundled fixtures —
no live stack, no network.

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

Runs every dynamic data-plane analyzer's built-in self-test and aggregates the result. Each analyzer must PASS its **golden-good** fixture AND FAIL its **golden-bad** fixture (the §11.4.107(10) self-validated-analyzer proof). An analyzer that passes its own golden-bad is a bluff gate, and this runner exits non-zero so it cannot ship.

This is the part of the dynamic harness that is **runnable immediately**: it validates analyzer correctness against the committed fixtures with zero network and no live dynamic stack.

Prerequisites

- bash/sh, awk, grep, tr on PATH.
- The six analyzers and their fixtures present under tests/dynamic/analyzers/ (each <signal>/fixtures/...).
- The committed tests/lib/evidence.sh (most analyzers delegate to it; a missing lib makes the delegating analyzers FAIL honestly, surfacing here as a self-test failure rather than a bluff).
- Write access to qa-results/ (gitignored) for the TAP artefact.

Usage examples

```
# Self-validate all six analyzers (golden-good PASS + golden-bad FAIL):
bash tests/dynamic/analyzers/run_analyzer_selftests.sh # exit 0
= all valid

# Inspect the aggregated TAP artefact:
cat qa-results/p9-harness/<run-id>/analyzer_selftests.tap
```

Edge cases

- **An analyzer passes its golden-bad fixture** → that analyzer's self-test exits non-zero, the runner marks it not ok, and the aggregate exits 1 (do NOT ship). This is the §1.1 paired-mutation behaviour for the analyzers.
- **An analyzer script is missing** → reported MISSING (not ok) and counted as a failure.
- **evidence.sh absent** → delegating analyzers FAIL their golden-good self-test (cannot delegate), surfacing as a self-test failure — never a silent pass.

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean (`sh -n + bash -n`, §11.4.67).
- Iterates the fixed list of six analyzers (`no_leak`, `graceful_503`, `egress_neq_host`, `xcache_hit`, `dns_no_plaintext_53`, `auth_407`), runs each with `--selftest`, and reuses the `analyzer_common.sh` TAP harness verdicts.
- Emits per-analyzer TAP plus an aggregate `1..N` plan and a `RESULT` line; tees everything to `qa-results/p9-harness/<run-id>/analyzer_selftests.tap`.
- Final exit: `grep -q '^not ok ' → exit 1`, else exit 0.

Related

- The six analyzers under `tests/dynamic/analyzers/*.sh` it drives.
- `tests/dynamic/lib/analyzer_common.sh` — supplies the self-test TAP harness.
- `tests/dynamic/parse_check.sh` — the sibling parseability gate.
- Constitution §11.4 / §11.4.69 / §11.4.107 / §1.1; design spec §13/§14.

Last verified

2026-07-01 — run against the bundled fixtures: all six analyzers self-validate (golden-good PASS + golden-bad FAIL); sh -n + bash -n parse-clean.